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Proc. R. Soc. Lond. A 1992 **437**, 311-327

doi: 10.1098/rspa.1992.0063

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Transformation field analysis of inelastic composite materials

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A new method is proposed for evaluation of local fields and overall properties of composite materials subjected to incremental thermomechanical loads and to transformation strains in the phases. The composite aggregate may consist of many perfectly bonded inelastic phases of arbitrary geometry and elastic material symmetry. In principle, any inviscid or time-dependent inelastic constitutive relation that complies with the additive decomposition of total strains can be admitted in the analysis. The governing system of equations is derived from the representation of local stress and strain fields by novel transformation influence functions and concentration factor tensors, as discussed in the preceding paper by G. J. Dvorak and Y. Benveniste. The concentration factors depend on local and overall thermoelastic moduli, and can be evaluated with a selected micromechanical model. Applications to elastic-plastic, viscoelastic, and viscoplastic systems are discussed. The new approach is contrasted with some presently accepted procedures based on the self-consistent and Mori-Tanaka approximations, which are shown to violate exact relations between local and overall inelastic strains.

1. Introduction

The purpose of this paper is to introduce a method for incremental micromechanical analysis of local fields and overall properties of inelastic heterogeneous media subjected to uniform thermomechanical loading along a prescribed path. The proposed approach relies on an explicit evaluation of piecewise uniform approximations of the residual fields that are introduced in multiphase solids by a distribution of piecewise uniform eigenstrains or eigenstresses, jointly referred to as transformation fields. As described in the preceding paper (I) by Dvorak & Benveniste (1992), such evaluations are made with the help of novel transformation influence functions, or concentration factor tensors. (References to, say, equation (23) in the companion paper (I) will be denoted here by (I 23), etc.) The treatment is suitable for composite aggregates made of any number of different, perfectly bonded inelastic phases which are represented by constitutive equations that admit the additive decomposition of total strains into elastic and inelastic components. However, any phase geometry and elastic material symmetry can be prescribed, and specific micromechanical models, such as the self-consistent, Mori-Tanaka, or unit cell approximations, are needed only in evaluations of the transformation concentration factor tensors. In any event, since the latter depend on local and overall thermoelastic moduli, only elastic solutions are required.

The elements of the present approach have been outlined by Dvorak (1991) for elastic plastic two-phase composite materials at small strains. The present work goes

Proc. R. Soc. Lond. A (1992) **437**, 311–327

Printed in Great Britain

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much farther in that it treats multiphase media and allows for many different types of inelastic behaviour. The opening §2 reviews the connections between the local and overall inelastic and total strains. The essence of the method is described in §3, through the formulation of governing systems of equations for evaluation of the total local strains or stresses, where the coefficients are formed, in part, by the transformation factor tensors. Section 4 is concerned with thermomechanical deformation of elastic-plastic composite systems. Explicit forms of the instantaneous mechanical and thermal concentration factors are found there for two and three-phase solids. Applications to viscoelastic and viscoplastic systems are described in §§5 and 6. Finally, §7 contrasts the new approach with some accepted procedures based on the self-consistent or Mori-Tanaka approximations, which are shown to violate a general connection between the local and overall inelastic strains.

2. Decompositions of local and overall fields

We are concerned with a representative volume V of a composite material that consists of many perfectly bonded phases which may have any physically admissible elastic symmetry and microstructural geometry. The volume V may be subdivided into several local volumes V_r , $r = 1, 2, \dots, N$, $\sum V_r = V$, such that each contains only one phase material, although any given phase may reside in more than one volume V_r . The boundary conditions imposed on V are limited to displacements compatible with a uniform overall strain $\boldsymbol{\varepsilon}$, or tractions derived from a uniform overall stress $\boldsymbol{\sigma}$, and a uniform temperature change θ ; all are expressed in a cartesian coordinate system $\mathbf{x}$ defined in V . The strains are assumed to be small, but both elastic and inelastic behaviour of the phases is admitted, providing that at each instant of loading it conforms with the additive decomposition

$$\boldsymbol{\sigma}_r(\mathbf{x}) = \boldsymbol{\sigma}_r^e(\mathbf{x}) + \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}), \quad \boldsymbol{\varepsilon}_r(\mathbf{x}) = \boldsymbol{\varepsilon}_r^e(\mathbf{x}) + \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}), \quad (1)$$

where the inelastic strain $\boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x})$ accumulates incrementally under applied stress, according to a certain phase constitutive relation which may or may not depend on time and temperature; specific examples of such relations will be discussed later. The $\boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x})$ represents a relaxation stress, that develops in a similar way under applied phase strain.

The $\boldsymbol{\sigma}_r^e(\mathbf{x})$ and $\boldsymbol{\varepsilon}_r^e(\mathbf{x})$ are elastic fields related by the usual constitutive relations, so that (1) can be recast as

$$\boldsymbol{\sigma}_r(\mathbf{x}) = \mathbf{L}_r \boldsymbol{\varepsilon}_r(\mathbf{x}) + \mathbf{l}_r \theta + \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}), \quad \boldsymbol{\varepsilon}_r(\mathbf{x}) = \mathbf{M}_r \boldsymbol{\sigma}_r(\mathbf{x}) + \mathbf{m}_r \theta + \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}), \quad (2)$$

in terms of the phase elastic stiffness $\mathbf{L}_r$, or compliance $\mathbf{M}_r = \mathbf{L}_r^{-1}$, and the thermal stress and strain tensors $\mathbf{l}_r$ and $\mathbf{m}_r = -\mathbf{M}_r \mathbf{l}_r$. Since θ and one of the inelastic fields are independent, it follows that

$$\mathbf{m}_r = -\mathbf{M}_r \mathbf{l}_r, \quad \mathbf{l}_r = -\mathbf{L}_r \mathbf{m}_r, \quad \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}) = -\mathbf{M}_r \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}), \quad \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}) = -\mathbf{L}_r \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}). \quad (3)$$

In the special case of a purely elastic response of the composite aggregate, the local fields and overall loads are related by the mechanical and thermal elastic influence functions

$$\left. \begin{aligned} \boldsymbol{\varepsilon}_r(\mathbf{x}) &= \mathbf{A}_r(\mathbf{x}) \boldsymbol{\varepsilon} + \mathbf{a}_r(\mathbf{x}) \theta & \text{if } \boldsymbol{\sigma}_s^{\text{re}}(\mathbf{x}) &= 0, \\ \boldsymbol{\sigma}_r(\mathbf{x}) &= \mathbf{B}_r(\mathbf{x}) \boldsymbol{\sigma} + \mathbf{b}_r(\mathbf{x}) \theta & \text{if } \boldsymbol{\varepsilon}_s^{\text{in}}(\mathbf{x}) &= 0. \end{aligned} \right\} s = 1, 2, \dots, r, \dots, N, \quad (4)$$

The total overall stress under an applied overall strain $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_0$, or the total overall

strain of the representative volume V under an overall stress $\boldsymbol{\sigma} = \boldsymbol{\sigma}_0$, and the temperature change θ_0 , are evaluated by integration of (2) over V . As in (1), one can recover the additive decompositions

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^e + \boldsymbol{\sigma}^{\text{re}}, \quad \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^{\text{in}}, \quad (5)$$

where the individual terms are defined as

$$\left. \begin{aligned} \boldsymbol{\sigma}^e &= \frac{1}{V} \int_V [\mathbf{L}_r \mathbf{A}_r(\mathbf{x}) \boldsymbol{\varepsilon}_0 + (\mathbf{L}_r \mathbf{a}_r(\mathbf{x}) + \mathbf{l}_r) \theta_0] dV, \\ \boldsymbol{\sigma}^{\text{re}} &= \frac{1}{V} \int_V [\mathbf{L}_r(\boldsymbol{\varepsilon}_r(\mathbf{x}) - \mathbf{A}_r(\mathbf{x}) \boldsymbol{\varepsilon}_0 - \mathbf{a}_r(\mathbf{x}) \theta_0) + \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x})] dV, \end{aligned} \right\} \quad (6)$$

$$\left. \begin{aligned} \boldsymbol{\varepsilon}^e &= \frac{1}{V} \int_V [\mathbf{M}_r \mathbf{B}_r(\mathbf{x}) \boldsymbol{\sigma}_0 + (\mathbf{M}_r \mathbf{b}_r(\mathbf{x}) + \mathbf{m}_r) \theta_0] dV, \\ \boldsymbol{\varepsilon}^{\text{in}} &= \frac{1}{V} \int_V [\mathbf{M}_r(\boldsymbol{\sigma}_r(\mathbf{x}) - \mathbf{B}_r(\mathbf{x}) \boldsymbol{\sigma}_0 - \mathbf{b}_r(\mathbf{x}) \theta_0) + \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x})] dV. \end{aligned} \right\} \quad (7)$$

The $\boldsymbol{\sigma}^e$ and $\boldsymbol{\varepsilon}^e$ represents the purely elastic, fully recoverable overall response to the applied loads $\boldsymbol{\varepsilon}_0$ or $\boldsymbol{\sigma}_0$, and θ_0 , that follows from (2) to (4); the associated local elastic fields in V are given by (4). The $\boldsymbol{\sigma}^{\text{re}}$ and $\boldsymbol{\varepsilon}^{\text{in}}$ are the overall stress and strain caused by the local inelastic fields which are, in general, independent of the current thermomechanical loads. However, to be admissible in the present analysis, the local inelastic strains and the corresponding residual fields must be associated with inelastic strains that are macroscopically uniform. Note that they are obtained by superposition of the volume averages of the local inelastic fields $\boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x})$ or $\boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x})$ themselves, with the residual elastic fields induced in the aggregate by those local fields, i.e. by $\boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x})$ at $\boldsymbol{\varepsilon} = \mathbf{0}$ and $\theta = 0$, or by $\boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x})$ at $\boldsymbol{\sigma} = \mathbf{0}$ and $\theta = 0$. The implication is that the inelastic fields, introduced by some thermomechanical loading history leading to the current values of $\boldsymbol{\varepsilon}$, $\boldsymbol{\sigma}$ and θ , cannot be recovered by an instantaneous elastic unloading.

The decomposition (5) of the overall response is consistent with the decomposition (1) of the total local strain or stress fields, in the sense that

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^e + \boldsymbol{\sigma}^{\text{re}} = \frac{1}{V} \int_V [\boldsymbol{\sigma}_r^e(\mathbf{x}) + \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x})] dV, \quad (8)$$

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^{\text{in}} = \frac{1}{V} \int_V [\boldsymbol{\varepsilon}_r^e(\mathbf{x}) + \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x})] dV. \quad (9)$$

However, the individual terms do not correspond to each other. On the local scale, the elastic fields in (1) results from the superposition of the elastic local fields (4) with the local residual elastic fields. On the overall scale, the elastic terms are volume integrals of the elastic fields (4), whereas the inelastic terms are volume integrals of the residual elastic fields superimposed with the inelastic local fields.

Evaluation of the inelastic terms in the above relations is facilitated by the results obtained in (I). In particular, if the local fields in (2) are compared with those in (I 2), and the overall fields in (8) with those in (I 1), the above thermal and inelastic fields are identified as certain local and overall transformation fields,

$$\boldsymbol{\lambda}_r(\mathbf{x}) = \mathbf{l}_r \theta + \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}), \quad \boldsymbol{\mu}_r(\mathbf{x}) = \mathbf{m}_r \theta + \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}), \quad (10)$$

$$\boldsymbol{\lambda} = \mathbf{l} \theta + \boldsymbol{\sigma}^{\text{re}}, \quad \boldsymbol{\mu} = \mathbf{m} \theta + \boldsymbol{\varepsilon}^{\text{in}}. \quad (11)$$

Since the thermal fields are uniform, one finds from (I 7) to (I 9) that

$$\mathbf{l} = \sum_{r=1}^N c_r \mathbf{A}_r^T \mathbf{l}_r, \quad \boldsymbol{\sigma}^{\text{re}} = \frac{1}{V} \int_V \mathbf{A}_r^T(\mathbf{x}) \boldsymbol{\sigma}_r^{\text{re}}(\mathbf{x}) dV, \quad (12)$$

$$\mathbf{m} = \sum_{r=1}^N c_r \mathbf{B}_r^T \mathbf{m}_r, \quad \boldsymbol{\varepsilon}^{\text{in}} = \frac{1}{V} \int_V \mathbf{B}_r^T(\mathbf{x}) \boldsymbol{\varepsilon}_r^{\text{in}}(\mathbf{x}) dV, \quad (13)$$

where $c_r = V_r/V$.

In the spirit of the approach outlined in §3 of (I), the actual local fields will be approximated below by piecewise uniform distributions. Under such circumstances all local fields and the influence functions in (2)–(7) are replaced by their averages over V_r , and one also recovers

$$\boldsymbol{\sigma}^{\text{re}} = \sum_{r=1}^N c_r \mathbf{A}_r^T \boldsymbol{\sigma}_r^{\text{re}}, \quad \boldsymbol{\varepsilon}^{\text{in}} = \sum_{r=1}^N c_r \mathbf{B}_r^T \boldsymbol{\varepsilon}_r^{\text{in}}. \quad (14)$$

It is then possible to evaluate the total stress and strain in (8) and (9) as

$$\boldsymbol{\sigma} = \mathbf{L}\boldsymbol{\varepsilon} + \mathbf{l}\theta + \sum_{r=1}^N c_r \mathbf{A}_r^T \boldsymbol{\sigma}_r^{\text{re}} = \sum_{r=1}^N c_r \boldsymbol{\sigma}_r, \quad (15)$$

$$\boldsymbol{\varepsilon} = \mathbf{M}\boldsymbol{\sigma} + \mathbf{m}\theta + \sum_{r=1}^N c_r \mathbf{B}_r^T \boldsymbol{\varepsilon}_r^{\text{in}} = \sum_{r=1}^N c_r \boldsymbol{\varepsilon}_r, \quad (16)$$

with the usual definitions of the overall elastic stiffness $\mathbf{L}$ and compliance $\mathbf{M} = \mathbf{L}^{-1}$, and the overall elastic thermal stress and strain tensors $\mathbf{l}$, $\mathbf{m}$,

$$\mathbf{M} = \sum_{r=1}^N c_r \mathbf{M}_r \mathbf{B}_r, \quad \mathbf{m} = \sum_{r=1}^N (\mathbf{M}_r \mathbf{b}_r + \mathbf{m}_r), \quad (17)$$

$$\mathbf{L} = \sum_{r=1}^N c_r \mathbf{L}_r \mathbf{A}_r, \quad \mathbf{l} = \sum_{r=1}^N (\mathbf{L}_r \mathbf{a}_r + \mathbf{l}_r), \quad (18)$$

that follows from (2)–(4), (6₁) and (7₁).

The expressions for overall averages are seen to be analogous to those for the local fields in (2), and one can also establish connections that are analogous to those in (3)

$$\mathbf{m} = -\mathbf{M}\mathbf{l}, \quad \mathbf{l} = -\mathbf{L}\mathbf{m}, \quad \boldsymbol{\varepsilon}^{\text{in}} = -\mathbf{M}\boldsymbol{\sigma}^{\text{re}}, \quad \boldsymbol{\sigma}^{\text{re}} = -\mathbf{L}\boldsymbol{\varepsilon}^{\text{in}}. \quad (19)$$

Note that in addition to the usual connections (15₂) and (16₂) between the local and overall total strains, there exist additional independent connections (14) between the local and overall inelastic fields. Sections 3–6 below outline the procedure that guarantees their satisfaction in various inelastic heterogeneous solids, while §7 points out that many micromechanical models currently in use violate (14).

Note also that in a homogeneous material ($\mathbf{L}_r = \mathbf{L}$, $\mathbf{l}_r = \mathbf{l}$, etc.), there is, according to (4), $\mathbf{A}_r(\mathbf{x}) = \mathbf{B}_r(\mathbf{x}) = \mathbf{I}$, and $\mathbf{a}_r(\mathbf{x}) = \mathbf{b}_r(\mathbf{x}) = \mathbf{0}$. Since the volume average of $\boldsymbol{\varepsilon}_r(\mathbf{x})$ in V is equal to $\boldsymbol{\varepsilon}_0$, and that of $\boldsymbol{\sigma}_r(\mathbf{x})$ to $\boldsymbol{\sigma}_0$, the volume integrals of the residual fields vanish and the overall inelastic terms are equal to the volume averages of the local inelastic fields.

3. Evaluation of the transformation fields

We now recall the representation of the local fields by the transformation concentration factor tensors defined in (I 13) and (I 14). For any representative volume under uniform overall strain $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_0$ or stress $\boldsymbol{\sigma} = \boldsymbol{\sigma}_0$, and a temperature

change $\theta = \theta_0$, which contains a piecewise uniform distribution of the thermal and inelastic fields associated with uniform overall loading and deformation, the averages of the local fields are written as

$$\boldsymbol{\varepsilon}_s = \mathbf{A}_s \boldsymbol{\varepsilon}_0 + \sum_{r=1}^N \mathbf{D}_{sr} (\mathbf{m}_r \theta_0 + \boldsymbol{\varepsilon}_r^{\text{in}}), \quad r, s = 1, 2, \dots, N, \quad (20)$$

$$\boldsymbol{\sigma}_s = \mathbf{B}_s \boldsymbol{\sigma}_0 + \sum_{r=1}^N \mathbf{F}_{sr} (\mathbf{l}_r \theta_0 + \boldsymbol{\sigma}_r^{\text{re}}), \quad r, s = 1, 2, \dots, N. \quad (21)$$

The $\mathbf{A}_s$ and $\mathbf{B}_s$ are the mechanical concentration factor tensors, and $\mathbf{D}_{sr}$, $\mathbf{F}_{sr}$ are certain eigenstrain and eigenstress concentration factor tensors. They all depend on the local and overall elastic moduli, and on the shape and volume fraction of the phases, and are therefore constant. The self-induced factors $\mathbf{D}_{ss}$ and $\mathbf{F}_{ss}$ contribute both the residual field caused in V_s by the transformation fields, and the fields themselves, these two contributions may be separated. Self-consistent and Mori–Tanaka estimates of these tensors were discussed in §6 in (I), the key results are reproduced in (63) and (64) below. Recall also that (I 76) provides the connections

$$\mathbf{a}_s = \sum_{r=1}^N \mathbf{D}_{sr} \mathbf{m}_r, \quad \mathbf{b}_s = \sum_{r=1}^N \mathbf{F}_{sr} \mathbf{l}_r, \quad (22)$$

hence the thermal terms in (20) and (21) may be eliminated from the sums.

Many inelastic constitutive laws relate either the local relaxation stress $\boldsymbol{\sigma}_r^{\text{re}}$ to the past history of the local strain $\boldsymbol{\varepsilon}_r$, or the strain $\boldsymbol{\varepsilon}_r^{\text{in}}$ to the history of the local stress $\boldsymbol{\sigma}_r$. When these stresses and strains are uniform in V_r , this can be formally written as

$$\boldsymbol{\sigma}_r^{\text{re}} = \mathbf{g}(\boldsymbol{\varepsilon}_r), \quad \boldsymbol{\varepsilon}_r^{\text{in}} = \mathbf{f}(\boldsymbol{\sigma}_r). \quad (23)$$

In some cases (e.g. in plasticity of metals) such relations exist between the respective increments. In any event, to accommodate (23), the equations (20) and (21) are modified by the identities $\boldsymbol{\varepsilon}_r^{\text{in}} = -\mathbf{M}_r \boldsymbol{\sigma}_r^{\text{re}}$ and $\boldsymbol{\sigma}_r^{\text{re}} = -\mathbf{L}_r \boldsymbol{\varepsilon}_r^{\text{in}}$, derived from (3), and by the relations (22). This provides the following two systems of governing equations for evaluation of the local strains and stresses

$$\boldsymbol{\varepsilon}_s + \mathbf{D}_{ss} \mathbf{M}_s \mathbf{g}(\boldsymbol{\varepsilon}_s) + \sum_{\substack{r=1 \\ r \neq s}}^N \mathbf{D}_{sr} \mathbf{M}_r \mathbf{g}(\boldsymbol{\varepsilon}_r) = \mathbf{A}_s \boldsymbol{\varepsilon} + \mathbf{a}_s \theta, \quad r, s = 1, 2, \dots, N, \quad (24)$$

$$\boldsymbol{\sigma}_s + \mathbf{F}_{ss} \mathbf{L}_s \mathbf{f}(\boldsymbol{\sigma}_s) + \sum_{\substack{r=1 \\ r \neq s}}^N \mathbf{F}_{sr} \mathbf{L}_r \mathbf{f}(\boldsymbol{\sigma}_r) = \mathbf{B}_s \boldsymbol{\sigma} + \mathbf{b}_s \theta, \quad r, s = 1, 2, \dots, N. \quad (25)$$

At any point of the prescribed overall thermomechanical loading path, these equations must be satisfied by the piecewise uniform approximations of the total local fields in the representative volume of an inelastic heterogeneous medium. As long as all the concentration factor tensors are constant, similar equations also hold for local and overall strain and stress increments, and for their time rates of change. If at least one of the above systems can be solved along the prescribed thermomechanical loading path, it yields piecewise uniform approximations to the total local fields in (2).

In this manner, the inelastic deformation problem for a multiphase composite material is reduced to evaluation of the mechanical, thermal and transformation

factor tensors, which depend only on the local and overall thermoelastic moduli, and to integration of one of the systems of the governing equations (24) or (25) along the prescribed loading path.

The concentration factors may be evaluated from a micromechanical model for elastic heterogeneous solids, as discussed in §§6 and 7 of (I). To be admissible, these factors must satisfy the general connections (I 47) to (I 50). This guarantees that the respective local and overall inelastic fields conform with the relations (12) to (14), and that the total local and overall stresses and strains satisfy (15) and (16). Therefore, one can substitute from (15₂) or (16₂) into (25) or (24), respectively, to reduce the number of unknowns in either system from N to $N-1$. Under certain circumstances, one can also use (15₁) or (16₁) to accomplish additional reductions; this is illustrated in §7 below.

In unit-cell models, the concentration factor tensors are usually found from elastic finite element solutions, and integration of (24) or (25) then reproduces the results that would be obtained from a finite element solution of the particular inelastic unit-cell domain during the prescribed loading history. Under certain circumstances, the present approach is more efficient than the finite element method, and in any case, it offers a particularly simple way for introduction of the constitutive relations of the phases.

The method of solution of (24) or (25) depends on the specific form of the inelastic constitutive equations (23), and is best illustrated by the examples that follow.

4. Elastic-plastic composite systems

As one of the possible applications of (24) and (25), we consider a composite aggregate with elastic-plastic phases (Dvorak 1991). No attempt will be made to spell out the details of the various constitutive theories. Instead, we adopt the general incremental form for material points which, subject to certain loading and unloading criteria, undergo plastic straining from some current state

$$d\sigma_r(\mathbf{x}) = \mathcal{L}_r[\epsilon_r(\mathbf{x}) - \beta_r(\mathbf{x}), H_r(\mathbf{x})] d\epsilon_r(\mathbf{x}) + \ell_r[H_r(\mathbf{x})] d\theta \quad (26)$$

$$d\epsilon_r(\mathbf{x}) = \mathcal{M}_r[\sigma_r(\mathbf{x}) - \alpha_r(\mathbf{x}), H_r(\mathbf{x})] d\sigma_r(\mathbf{x}) + m_r[H_r(\mathbf{x})] d\theta. \quad (27)$$

The $\mathcal{L}_r$ and $\mathcal{M}_r$ are the instantaneous mechanical stiffness and compliance, and ℓ_r , m_r are the instantaneous thermal stress and strain vectors which usually reflect the variation of yield stress with temperature. The above $\mathcal{L}_r$, $\mathcal{M}_r$ depend on the magnitude of the invariants of the local strain $\epsilon_r(\mathbf{x}) - \beta_r(\mathbf{x})$, or stress $\sigma_r(\mathbf{x}) - \alpha_r(\mathbf{x})$, where $\beta_r(\mathbf{x})$ and $\alpha_r(\mathbf{x})$ denote certain back-strain or stress terms, associated with the centres of the current relaxation and yield surface of the material at point $\mathbf{x}$ in V_r ; $H_r(\mathbf{x})$ is a functional of past deformation history.

Volume averages of (26) and (27) can be evaluated only if the actual fields are known. In piecewise uniform approximations of those fields, one may replace (26) and (27) with

$$d\sigma_r \doteq \mathcal{L}_r(\epsilon_r - \beta_r, H_r) d\epsilon_r + \ell_r(H_r) d\theta, \quad d\epsilon_r \doteq \mathcal{M}_r(\sigma_r - \alpha_r, H_r) d\sigma_r + m_r(H_r) d\theta, \quad (28)$$

in each local volume V_r . Of course, if the subdivision of the representative volume V into local volumes V_r is such that the actual fields in V_r deviate substantially from their respective averages, this replacement may lead to large errors.

From (26) and (2), the local relaxation stress and plastic strain are,

$$d\sigma_r^{\text{re}} = (\mathcal{L}_r - L_r) d\epsilon_r + (\ell_r - l_r) d\theta, \quad d\epsilon_r^{\text{in}} = (\mathcal{M}_r - M_r) d\sigma_r + (m_r - m_r) d\theta, \quad (29)$$

or, in a condensed version, with instantaneous plastic stiffness and compliance tensors

$$d\sigma_r^e = \mathcal{L}_r^p d\epsilon_r + \ell_r^p d\theta, \quad d\sigma_r^{\text{in}} = \mathcal{M}_r^p d\sigma_r + m_r^p d\theta. \quad (30)$$

When this is substituted into the incremental form of (24) and (25), the governing equations for evaluation of the local strain and stress increments become

$$d\epsilon_s + \sum_{r=1}^N \mathbf{D}_{sr} \mathbf{M}_r \mathcal{L}_r^p d\epsilon_r = \mathbf{A}_s d\epsilon + \left(\mathbf{a}_s - \sum_{r=1}^N \mathbf{D}_{sr} \mathbf{M}_r \ell_r^p \right) d\theta, \quad (31)$$

$$d\sigma_s + \sum_{r=1}^N \mathbf{F}_{sr} \mathbf{L}_r \mathcal{M}_r^p d\sigma_r = \mathbf{B}_s d\sigma + \left(\mathbf{b}_s - \sum_{r=1}^N \mathbf{F}_{sr} \mathbf{L}_r m_r^p \right) d\theta. \quad (32)$$

In multiphase systems, these equations are best solved numerically, but closed-form solutions can be easily obtained, for example, for systems consisting of two or three phases. In any case, it is desirable to write the result in the form that is analogous to (4),

$$d\epsilon_r = \mathcal{A}_r d\epsilon + a_r d\theta, \quad d\sigma_r = \mathcal{B}_r d\sigma + \ell_r d\theta, \quad (33)$$

where $\mathcal{A}_r$, a_r and $\mathcal{B}_r$, ℓ_r are the instantaneous mechanical and thermal strain and stress concentration factor tensors for the local volumes V_r .

Once the instantaneous concentration factor tensors are known, the instantaneous overall stiffness and compliance of the inelastic composite medium can be defined as tensors relating the overall stress and strain at any instant of loading. The derivation and the resulting expressions are entirely analogous to the elastic case cf. (15) to (18),

$$d\sigma = \mathcal{L} d\epsilon + \ell d\theta, \quad d\epsilon = \mathcal{M} d\sigma + m d\theta, \quad (34)$$

$$\mathcal{L} = \sum_{r=1}^N c_r \mathcal{L}_r \mathcal{A}_r, \quad \ell = \sum_{r=1}^N c_r (\mathcal{L}_r a_r + \ell_r), \quad (35)$$

$$\mathcal{M} = \sum_{r=1}^N c_r \mathcal{M}_r \mathcal{B}_r, \quad m = \sum_{r=1}^N c_r (\mathcal{M}_r \ell_r + m_r). \quad (36)$$

As one illustration, we present the closed-form solution for a *two-phase system*, where the phases are denoted as $r = \alpha, \beta$, and $s = \alpha, \beta$. Then, (31) or (32) each provide two equations, which may be supplemented by (15₂) and (16₂), solved in the form (33), and thus give the concentration factor tensors

$$\left. \begin{aligned} \mathcal{A}_\alpha &= [\mathbf{I} + \mathbf{D}_{\alpha\alpha} \mathbf{M}_\alpha \mathcal{L}_\alpha^p - (c_\alpha/c_\beta) \mathbf{D}_{\alpha\beta} \mathbf{M}_\beta \mathcal{L}_\beta^p]^{-1} [\mathbf{A}_\alpha - (1/c_\beta) \mathbf{D}_{\alpha\beta} \mathbf{M}_\beta \mathcal{L}_\beta^p], \\ a_\alpha &= [\mathbf{I} + \mathbf{D}_{\alpha\alpha} \mathbf{M}_\alpha \mathcal{L}_\alpha^p - (c_\alpha/c_\beta) \mathbf{D}_{\alpha\beta} \mathbf{M}_\beta \mathcal{L}_\beta^p]^{-1} [\mathbf{a}_\alpha - \mathbf{D}_{\alpha\alpha} \mathbf{M}_\alpha \ell_\alpha^p - \mathbf{D}_{\alpha\beta} \mathbf{M}_\beta \ell_\beta^p]. \end{aligned} \right\} \quad (37)$$

$$\left. \begin{aligned} \mathcal{B}_\alpha &= [\mathbf{I} + \mathbf{F}_{\alpha\alpha} \mathbf{L}_\alpha \mathcal{M}_\alpha^p - (c_\alpha/c_\beta) \mathbf{F}_{\alpha\beta} \mathbf{L}_\beta \mathcal{M}_\beta^p]^{-1} [\mathbf{B}_\alpha - (1/c_\beta) \mathbf{F}_{\alpha\beta} \mathbf{L}_\beta \mathcal{M}_\beta^p], \\ \ell_\alpha &= [\mathbf{I} + \mathbf{F}_{\alpha\alpha} \mathbf{L}_\alpha \mathcal{M}_\alpha^p - (c_\alpha/c_\beta) \mathbf{F}_{\alpha\beta} \mathbf{L}_\beta \mathcal{M}_\beta^p]^{-1} [\mathbf{b}_\alpha - \mathbf{F}_{\alpha\alpha} \mathbf{L}_\alpha m_\alpha^p - \mathbf{F}_{\alpha\beta} \mathbf{L}_\beta m_\beta^p]. \end{aligned} \right\} \quad (38)$$

The concentration factors for phase β are obtained by exchange of the α and β subscripts.

One may recall here that the transformation concentration factor tensors of two-phase materials are related by exact connections to the elastic mechanical concentration factors (Dvorak 1990, equations (123)–(126))

$$\left. \begin{aligned} \mathbf{D}_{r\alpha} &= (\mathbf{I} - \mathbf{A}_r) (\mathbf{L}_\alpha - \mathbf{L}_\beta)^{-1} \mathbf{L}_\alpha, & \mathbf{D}_{r\beta} &= -(\mathbf{I} - \mathbf{A}_r) (\mathbf{L}_\alpha - \mathbf{L}_\beta)^{-1} \mathbf{L}_\beta, \\ \mathbf{F}_{r\alpha} &= (\mathbf{I} - \mathbf{B}_r) (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathbf{M}_\alpha, & \mathbf{F}_{r\beta} &= -(\mathbf{I} - \mathbf{B}_r) (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathbf{M}_\beta. \end{aligned} \right\} \quad (39)$$

These may be substituted into (37) and (38) to derive expressions for the instantaneous concentration factor tensors which depend only on the elastic

mechanical concentration factor tensors, phase volume fractions, and on the instantaneous phase properties. If desired, the dependence of the instantaneous concentration factor tensors on local geometry can be eliminated altogether by appealing to (A 17) in the Appendix, which relates the elastic concentration factor tensors to the overall elastic moduli $\mathbf{L}$ or compliances $\mathbf{M}$.

Another illustration pertains to a *three-phase* system, $r = 1, 2, 3$, where one may write (31) or (32), together with (15₂) or (16₂), to express one of the unknown strains, say $d\mathbf{e}_3$, as

$$d\mathbf{e}_3 = (d\mathbf{e} - c_1 d\mathbf{e}_1 - c_2 d\mathbf{e}_2)/c_3. \quad (40)$$

This reduces the system (31) to two equations that can be readily solved. With reference to (33₁), one finds the following instantaneous strain concentration factors,

$$\left. \begin{aligned} \mathcal{A}_1 &= [\mathbf{Z}_2^{-1} \mathbf{Z}_1 - \mathbf{Y}_2^{-1} \mathbf{Y}_1]^{-1} [\mathbf{Z}_2^{-1} \mathbf{Z}_3 - \mathbf{Y}_2^{-1} \mathbf{Y}_3]^{-1}, \\ a_1 &= [\mathbf{Z}_2^{-1} \mathbf{Z}_1 - \mathbf{Y}_2^{-1} \mathbf{Y}_1]^{-1} [\mathbf{Z}_2^{-1} \mathbf{z}_4 - \mathbf{Y}_2^{-1} \mathbf{y}_4], \end{aligned} \right\} \quad (41)$$

$$\left. \begin{aligned} \mathcal{A}_2 &= [\mathbf{Z}_1^{-1} \mathbf{Z}_2 - \mathbf{Y}_1^{-1} \mathbf{Y}_2]^{-1} [\mathbf{Z}_1^{-1} \mathbf{Z}_3 - \mathbf{Y}_1^{-1} \mathbf{Y}_3], \\ a_2 &= [\mathbf{Z}_2^{-1} \mathbf{Z}_1 - \mathbf{Y}_2^{-1} \mathbf{Y}_1]^{-1} [\mathbf{Z}_1^{-1} \mathbf{z}_4 - \mathbf{Y}_1^{-1} \mathbf{y}_4], \end{aligned} \right\} \quad (42)$$

$$\mathcal{A}_3 = (1/c_3) [\mathbf{I} - c_1 \mathcal{A}_1 - c_2 \mathcal{A}_2], \quad a_3 = (1/c_3) [\mathbf{I} - c_1 a_1 - c_2 a_2], \quad (43)$$

where the auxiliary tensors were defined as

$$\left. \begin{aligned} \mathbf{Z}_1 &= \mathbf{I} + \mathbf{D}_{11} \mathbf{M}_1 \mathcal{L}_1^p - (c_1/c_3) \mathbf{D}_{13} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{Z}_2 &= \mathbf{D}_{12} \mathbf{M}_2 \mathcal{L}_2^p - (c_2/c_3) \mathbf{D}_{13} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{Z}_3 &= \mathbf{A}_1 - (1/c_3) \mathbf{D}_{13} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{z}_4 &= \mathbf{a}_1 - \mathbf{D}_{11} \mathbf{M}_1 \mathcal{L}_1^p - \mathbf{D}_{12} \mathbf{M}_2 \mathcal{L}_2^p - \mathbf{D}_{13} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{Y}_1 &= \mathbf{D}_{21} \mathbf{M}_1 \mathcal{L}_1^p - (c_1/c_3) \mathbf{D}_{23} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{Y}_2 &= \mathbf{I} + \mathbf{D}_{22} \mathbf{M}_2 \mathcal{L}_2^p - (c_2/c_3) \mathbf{D}_{23} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{Y}_3 &= \mathbf{A}_2 - (1/c_3) \mathbf{D}_{23} \mathbf{M}_3 \mathcal{L}_3^p, \\ \mathbf{y}_4 &= \mathbf{a}_2 - \mathbf{D}_{21} \mathbf{M}_1 \mathcal{L}_1^p - \mathbf{D}_{22} \mathbf{M}_2 \mathcal{L}_2^p - \mathbf{D}_{23} \mathbf{M}_3 \mathcal{L}_3^p. \end{aligned} \right\} \quad (44)$$

5. Linearly viscoelastic composite systems

The typical approach to problems of this kind uses the correspondence principle of linear viscoelasticity, to relate the effective viscoelastic properties to the effective elastic properties. In general, an exact analytical solution of an elasticity problem for a given geometry of a heterogeneous medium can be converted into a transform parameter multiplied Laplace or Fourier transform of an analogous viscoelasticity problem. The latter problem is thus reduced to an inversion of the respective transformed solution (Christensen 1971).

The present approach offers an alternative which may be useful when the analytic elasticity solution or the inversion prove difficult to find. As before, we consider a representative volume of the composite aggregate, but allow viscoelastic deformation to take place in one or more phases. The total strain in each local volume is assumed to conform with the additive decomposition (1), where the elastic fields correspond to an instantaneous elastic response and thus depend only on the current local strain or stress according to (2), while the inelastic fields are certain functions of time and of the local strain or stress history. The overall stress or strain and temperature change applied to the representative volume are assumed to be uniform. The

derivation is limited to thermorheologically simple materials, where the effect of a uniform temperature change on the relaxation function is reflected by replacing the time variable in this function by a new variable that depends both on time and on a temperature-dependent shift function. Therefore, it is sufficient to consider isothermal viscoelastic behaviour in the derivations that follow, and allow for the influence of temperature changes on these relations only in actual evaluations.

In linear viscoelastic solids, the total strain caused by a history of applied stress that starts from zero stress at $t = 0$, is usually described by (Christensen 1971)

$$\varepsilon_r(t) = \int_0^t J_r(t-\tau) \frac{d\sigma_r(\tau)}{d\tau} d\tau, \quad (45)$$

where the function $J_r(t-\tau)$ is the creep compliance, assumed to possess the symmetries $J_{ijkl}^r = J_{jikl}^r = J_{klji}^r$. If the strains due to the instantaneous elastic response can be separated from the total strain (45), as in (2₂), the corresponding elastic moduli serve in evaluation of the various concentration factor tensors in (24) and (25), while the inelastic part of (45) is substituted into (23) and then into (24).

In a similar manner, if the local volume V_r is subjected to a certain time history of prescribed deformation starting from $\varepsilon_r = \mathbf{0}$ at $t = 0$, one can write the resulting total stress as

$$\sigma_r(t) = \int_0^t G_r(t-\tau) \frac{d\varepsilon_r(\tau)}{d\tau} d\tau, \quad (46)$$

where $G_r(t-\tau)$ is the local relaxation function which has the same symmetries as the creep compliance, and then call upon (2₁) to separate the instantaneous elastic response. Again, the resulting elastic moduli are used to evaluate the concentration factor tensors that appear in (25), and the relaxation stress component of (46) is substituted into (23) and (25). In either case, one obtains a system of integral equations for evaluation of the total local strains or stresses.

The preferred form of the solution of (24) or (25) for the local stresses and strains at a given time t of the overall loading history is

$$\varepsilon_r(t) = \mathfrak{A}_r(t) \varepsilon(t) + \alpha_r(t) \theta(t), \quad \sigma_r(t) = \mathfrak{B}_r(t) \sigma(t) + \mathfrak{b}_r(t) \theta(t), \quad (47)$$

where $\mathfrak{A}_r(t)$, $\alpha_r(t)$, and $\mathfrak{B}_r(t)$, $\mathfrak{b}_r(t)$ are certain mechanical and thermal influence functions corresponding to the prescribed history of a uniform overall strain or stress. As in (15) and (16), these results may be used to evaluate the total overall stress or strain at time t .

In addition to the total local and overall strains and stresses, it may be necessary to find the corresponding rates at time t . As long as the instantaneous elastic moduli used in evaluation of the concentration factor tensors that enter (24) and (25) remain constant, one can rewrite these equations for the local and overall rates. The corresponding constitutive relations for the rates are the time derivatives of (45) and (46). For example, if one considers a local deformation history starting from $\sigma_r = \mathbf{0}$ at $t = 0$, (45) can be differentiated with respect to time, and the result integrated by parts to yield

$$\dot{\varepsilon}_r(t) = \mathbf{M}_r \dot{\sigma}_r(t) + \dot{J}_r(0) \sigma_r(t) + \int_0^t \ddot{J}_r(t-\tau) \sigma_r(\tau) d\tau. \quad (48)$$

The elastic part of the total strain rate, $\dot{\varepsilon}_r^e(t) = \mathbf{M}_r \dot{\sigma}_r(t)$, replaces here the original term $J_r(0) \dot{\sigma}_r(t)$, that reflects the instantaneous elastic response.

Similarly, for a local deformation history that starts from $\boldsymbol{\varepsilon} = \mathbf{0}$ at $t = 0$, (46) can be differentiated and then integrated by parts to establish that the total local stress rate is

$$\dot{\boldsymbol{\sigma}}_r(t) = \mathbf{L}_r \dot{\boldsymbol{\varepsilon}}_r(t) + \dot{\mathbf{G}}_r(0) \boldsymbol{\varepsilon}_r(t) + \int_0^t \ddot{\mathbf{G}}_r(t-\tau) \boldsymbol{\varepsilon}_r(\tau) d\tau, \quad (49)$$

where the elastic part of the total stress rate, $\dot{\boldsymbol{\sigma}}_r^e(t) = \mathbf{L}_r \dot{\boldsymbol{\varepsilon}}_r(t) = \mathbf{G}_r(0) \dot{\boldsymbol{\varepsilon}}_r(t)$.

If the solution of the equations (24) or (25) for the respective rates is found in the form

$$\dot{\boldsymbol{\varepsilon}}_r(t) = \mathcal{A}_r(t) \dot{\boldsymbol{\varepsilon}}(t) + a_r(t) \dot{\theta}(t), \quad \dot{\boldsymbol{\sigma}}_r(t) = \mathcal{B}_r(t) \dot{\boldsymbol{\sigma}}(t) + \ell_r(t) \dot{\theta}(t), \quad (50)$$

one can then define the instantaneous overall stiffness and compliance, and the thermal strain and stress tensors of the composite medium as

$$\dot{\boldsymbol{\sigma}}(t) = \mathcal{L}(t) \dot{\boldsymbol{\varepsilon}}(t) + \ell(t) \dot{\theta}(t), \quad \dot{\boldsymbol{\varepsilon}}(t) = \mathcal{M}(t) \dot{\boldsymbol{\sigma}}(t) + m(t) \dot{\theta}(t), \quad (51)$$

and evaluate them as suggested by (35) and (36). The $\mathcal{A}_r(t)$ and $\mathcal{B}_r(t)$ in (50) are the instantaneous mechanical concentration factor tensors, and the $a_r(t)$ and $\ell_r(t)$ are the instantaneous thermal concentration factor tensors. The latter may reflect both the elastic response and any contribution that a temperature change may make to the inelastic terms in (45) to (49), under the stated assumption of thermorheologically simple phase materials.

Once again, the problem can be simplified if the composite aggregate under consideration consists of only two phases and local volumes $r = \alpha, \beta$. Then, one can employ (15₂) and (16₂) to write the connections $c_\alpha \boldsymbol{\sigma}_\alpha(t) + c_\beta \boldsymbol{\sigma}_\beta(t) = \boldsymbol{\sigma}(t)$, $c_\alpha \boldsymbol{\varepsilon}_\alpha(t) + c_\beta \boldsymbol{\varepsilon}_\beta(t) = \boldsymbol{\varepsilon}(t)$, and recover separate equations for each of the unknowns.

For example, consider a two-phase composite system subjected to a prescribed history of overall strain $\boldsymbol{\varepsilon}(t)$ and uniform temperature change $\theta(t)$. Let the phase constitutive relations be known in the form (45), or (46). To evaluate the estimates of local strain rates at time t , in terms of a piecewise uniform distribution in the two phases, appeal to the rate form of (24) and write the governing equations for this case as

$$\left. \begin{aligned} \dot{\boldsymbol{\varepsilon}}_\alpha(t) + \mathbf{D}_{\alpha\alpha} \dot{\boldsymbol{\sigma}}_\alpha^{\text{re}}(t) + \mathbf{D}_{\alpha\beta} \mathbf{M}_\beta \dot{\boldsymbol{\sigma}}_\beta^{\text{re}}(t) &= \mathbf{A}_\alpha \dot{\boldsymbol{\varepsilon}}(t) + \mathbf{a}_\alpha \dot{\theta}(t), \\ \dot{\boldsymbol{\varepsilon}}_\beta(t) + \mathbf{D}_{\beta\alpha} \mathbf{M}_\alpha \dot{\boldsymbol{\sigma}}_\alpha^{\text{re}}(t) + \mathbf{D}_{\beta\beta} \mathbf{M}_\beta \dot{\boldsymbol{\sigma}}_\beta^{\text{re}}(t) &= \mathbf{A}_\beta \dot{\boldsymbol{\varepsilon}}(t) + \mathbf{a}_\beta \dot{\theta}(t). \end{aligned} \right\} \quad (52)$$

Next, let the relaxation stresses be expressed by the last two terms in (49). To separate the variables, substitute in turn for one of the local strain rates from the above connections, to recover two uncoupled equations for the two rates. Only one of them needs to be considered; for example the equation for evaluation of the rate $\dot{\boldsymbol{\varepsilon}}_\beta(t)$ is

$$\begin{aligned} [(c_\beta/c_\alpha) \mathbf{D}_{\alpha\alpha}^{-1} + \mathbf{D}_{\beta\alpha}^{-1}] \dot{\boldsymbol{\varepsilon}}_\beta(t) &= (\mathbf{D}_{\alpha\alpha}^{-1} \mathbf{D}_{\alpha\beta} - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{D}_{\beta\beta}) \mathbf{M}_\beta \left[\dot{\mathbf{G}}_\beta(0) \boldsymbol{\varepsilon}_\beta(t) + \int_0^t \ddot{\mathbf{G}}_\beta(t-\tau) \boldsymbol{\varepsilon}_\beta(\tau) d\tau \right] \\ &\quad - [\mathbf{D}_{\alpha\alpha}^{-1} (\mathbf{A}_\alpha - c_\alpha^{-1} \mathbf{I}) - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{A}_\beta] \dot{\boldsymbol{\varepsilon}}(t) - (\mathbf{D}_{\alpha\alpha}^{-1} \mathbf{a}_\alpha - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{a}_\beta) \dot{\theta}(t), \end{aligned} \quad (53)$$

where the concentration factors $\mathbf{A}_\alpha$, $\mathbf{a}_\alpha$, $\mathbf{D}_{\alpha\beta}$, etc., are evaluated from the instantaneous elastic moduli. Once this has been solved, one can find the other rate as $\dot{\boldsymbol{\varepsilon}}_\alpha(t) = (\dot{\boldsymbol{\varepsilon}}(t) - c_\beta \dot{\boldsymbol{\varepsilon}}_\beta(t))/c_\alpha$, and add the resulting increments in total strains to the current values.

A similar procedure can be followed in evaluation of the total local strains. In

particular, for the above two-phase system subjected to the prescribed history of overall strain $\boldsymbol{\varepsilon}(t)$ and uniform temperature change $\theta(t)$, (24) provides two equations for the local strain fields which can be solved for one of the unknowns, say $\boldsymbol{\varepsilon}_\beta(t)$, to yield

$$\begin{aligned} & [(c_\beta/c_\alpha) \mathbf{D}_{\alpha\alpha}^{-1} + \mathbf{D}_{\beta\alpha}^{-1} + (\mathbf{D}_{\alpha\alpha}^{-1} \mathbf{D}_{\alpha\beta} - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{D}_{\beta\beta}) \mathbf{M}_\beta \mathbf{G}_\beta(0)] \boldsymbol{\varepsilon}(t) \\ &= (\mathbf{D}_{\alpha\alpha}^{-1} \mathbf{D}_{\alpha\beta} - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{D}_{\beta\beta}) \mathbf{M}_\beta \int_0^t \mathbf{G}_\beta(t-\tau) \dot{\boldsymbol{\varepsilon}}_\beta(\tau) d\tau \\ &- [\mathbf{D}_{\alpha\alpha}^{-1} (\mathbf{A}_\alpha - c_\alpha^{-1} \mathbf{I}) - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{A}_\beta] \boldsymbol{\varepsilon}(t) - (\mathbf{D}_{\alpha\alpha}^{-1} \mathbf{a}_\alpha - \mathbf{D}_{\beta\alpha}^{-1} \mathbf{a}_\beta) \theta(t). \end{aligned} \quad (54)$$

6. Viscoplastic systems

As pointed out in a recent review by Chaboche (1989), most constitutive theories for viscoplastic deformation of metals conform with the additive decomposition (1), and are of the unified type, i.e. no distinction is made between the inviscid and viscous part of the inelastic strain. The general framework of the unified theories often involves an assumed or implied viscoplastic potential Ω . The inelastic strain rate is then expressed, with reference to a particular homogeneous phase r , as

$$\dot{\boldsymbol{\varepsilon}}_r^{\text{in}} = \partial \Omega_r / \partial \boldsymbol{\sigma}_r = \frac{3}{2} \langle \sigma_r^v / K_r^* \rangle^n (\boldsymbol{\sigma}_r' - \mathbf{X}_r) / \mathbf{J}(\boldsymbol{\sigma}_r - \mathbf{X}_r), \quad (55)$$

with the overstress or viscous stress σ_r^v and the function $\mathbf{J}(\boldsymbol{\sigma}_r - \mathbf{X}_r)$ defined by

$$\sigma_r^v = \mathbf{J}(\boldsymbol{\sigma}_r - \mathbf{X}_r) - R_r^* - k_r^*, \quad \mathbf{J}(\boldsymbol{\sigma}_r - \mathbf{X}_r) = [3(\boldsymbol{\sigma}_r' - \mathbf{X}_r')(\boldsymbol{\sigma}_r' - \mathbf{X}_r')/2]^{1/2}, \quad (56)$$

where $\mathbf{X}_r$ is the back stress; $\boldsymbol{\sigma}_r'$, $\mathbf{X}_r'$ denote the deviators of $\boldsymbol{\sigma}_r$ and $\mathbf{X}_r$. The scalar k_r^* denotes the initial yield stress magnitude and R_r^* its evolution, their sum is the equilibrium stress corresponding to a vanishing strain rate at constant stress; K_r^* is the drag stress; each may be a function of temperature. The bracket $\langle x \rangle = xH(x)$, where $H(x)$ is the Heaviside function.

Several particular forms and variants of (55) are in use, with different time and temperature-dependent evolution rules for the variables $\mathbf{X}$, K^* , and R^* , or for analogous variables. Some theories (Krempel & Lu 1984) forego the assumption of a viscoplastic potential, and the associated loading/unloading criteria, and regard the entire deformation process as rate dependent, even at vanishingly small rates. This can be convenient in applications which are likely to lead to complex loading histories with frequent load reversals. In any case, actual evaluations of the inelastic strains for a prescribed stress history tend to be quite involved, and are often predicated on the availability of specific magnitudes of many material parameters. Best agreement between predicted and actual behaviour usually obtains under cyclic loading, where the requisite material parameters appear to have more reproducible magnitudes.

If (55) or its equivalent is taken to represent the response of one or more phases of a composite material, the local inelastic strain is obtained by time integration of (55), under the actual local stress history. The latter is not known *a priori*, but may be evaluated at any time t , from the solution of the system (25), rewritten as

$$\boldsymbol{\sigma}_s(t) + \sum_{r=1}^N \mathbf{F}_{sr} \mathbf{L}_r \int_0^t \dot{\boldsymbol{\varepsilon}}_r^{\text{in}}(\tau) d\tau = \mathbf{B}_s \boldsymbol{\sigma}(t) + \mathbf{b}_s \theta(t), \quad r, s = 1, 2, \dots, N, \quad (57)$$

under the initial conditions $\boldsymbol{\sigma}_r = \mathbf{0}$, $\theta = 0$ at $t = 0$, boundary conditions $\boldsymbol{\sigma} = \boldsymbol{\sigma}(t)$, and

a uniform temperature change $\theta = \theta(t)$ in the representative volume. The preferred form of the solution is formally identical with that given by (47₂). For a two-phase system, one can easily write a governing equation similar to (54).

Equation (57) can also be used to analyse systems where the total strain rate of one or more phases is described by nonlinear functions of the current stress, providing that the total strain can be decomposed according to (1). This may be useful in nonlinear viscous systems with phases undergoing power-law creep.

7. Comparison with related methods

The standard procedure for evaluation of the overall instantaneous properties of inelastic heterogeneous media is based on the early work by Hill (1965, 1966, 1967), and its elaboration by Hutchinson (1970) and by many later writers. The basic idea is to estimate the total strains or stresses in the phases, under an applied history of uniform overall strain or stress, by the self-consistent method. For example, in an elastic–plastic, isothermally deformed composite or polycrystal one can rewrite (33) as

$$d\epsilon_r = \mathcal{A}_r d\epsilon, \quad d\sigma_r = \mathcal{B}_r d\sigma. \quad (58)$$

The self-consistent estimates of the instantaneous concentration factor tensors are found from the solution of an inclusion problem, in which an ellipsoidal volume of each phase is embedded in an elastic homogeneous medium with certain instantaneous stiffness $\mathcal{L}$, or compliance $\mathcal{M}$. The $\mathcal{L}$ and $\mathcal{M}$ are identified with the as yet unknown effective properties of the composite aggregate, and the concentration factor tensors are found from the formula (I 54), written as

$$\mathcal{A}_r = (\mathcal{L}^* + \mathcal{L}_r)^{-1}(\mathcal{L}^* + \mathcal{L}), \quad \mathcal{B}_r = (\mathcal{M}^* + \mathcal{M}_r)^{-1}(\mathcal{M}^* + \mathcal{M}). \quad (59)$$

Here, the $\mathcal{L}_r$ and $\mathcal{M}_r$ are the instantaneous phase stiffness and compliance matrices defined by (28), and the $\mathcal{L}^*$ and $\mathcal{M}^*$ are the instantaneous constraint tensors of the transformed homogeneous inclusion in the effective medium.

The unknown instantaneous overall properties are found from the general connections (15₂) and (16₂), in the form analogous to (34), (35₁), and (36₁),

$$\mathcal{L} = \sum_{r=1}^N c_r \mathcal{L}_r \mathcal{A}_r, \quad \mathcal{M} = \sum_{r=1}^N c_r \mathcal{M}_r \mathcal{B}_r. \quad (60)$$

In this manner, the incremental solution of an isothermal loading problem for an elastic–plastic composite is reduced to a sequence of elasticity problems for a composite with varying local and overall moduli. We recall from (I 53) to (I 57) that approximate solutions of such elasticity problems by the self-consistent method are formally similar to solutions by the Mori–Tanaka method, providing that the constraint tensors in (59) are evaluated from the instantaneous properties of the matrix. Therefore, with this proviso, the above relations can also be used to construct the Mori–Tanaka estimates of instantaneous overall properties of the inelastic composite. Such results were recently found from a different approach by Tandon & Weng (1988), Gavazzi & Lagoudas (1990) and others. Of course, to assure that these expressions can be reduced to a diagonally symmetric form, cf. (I 55), it is necessary to limit the selection of admissible heterogeneous aggregates either to two-phase composites, or to multiphase systems reinforced by inclusions of the same shape and alignment, or to multiphase systems reinforced by similarly shaped but randomly orientated inclusions (Benveniste *et al.* 1991; Chen *et al.* 1992).

It is clear that the above estimates (60) of the instantaneous $\mathcal{L}$ and $\mathcal{M}$ tensors do comply with the general connections (15₂) and (16₂) between the local and overall total strains and stresses. If the elastic compliance is subtracted from the total, one finds the overall instantaneous inelastic compliance as

$$\mathcal{M} - \mathbf{M} = \sum_{r=1}^N c_r (\mathcal{M}_r \mathcal{B}_r - \mathbf{M}_r \mathbf{B}_r). \quad (61)$$

However, if one substitutes into (14₂) the $d\mathbf{e}_r^{\text{in}}$ from (29₂) for $d\theta = 0$, and uses (33) to replace the local stress $d\boldsymbol{\sigma}_r$ by the overall stress $d\boldsymbol{\sigma}$, then (16₁) can be used to find the overall inelastic compliance as

$$\mathcal{M} - \mathbf{M} = \sum_{r=1}^N c_r \mathbf{B}_r^T (\mathcal{M}_r - \mathbf{M}_r) \mathcal{B}_r. \quad (62)$$

These are completely general expressions that follow from the representation of local stresses by (4) and (58), and from the two independent connections between the local and overall inelastic strains (14₂) and total strains (16₂), respectively. It is interesting to note that (61) and (62) may also be derived as two different self-consistent estimates of $\mathcal{M}$. Of course, the first equation (61) is such an estimate for a multiphase solid where the overall total strain complies with the condition $\boldsymbol{\varepsilon} = \sum c_r \boldsymbol{\varepsilon}_r$. In contrast, (62) is a similar estimate of $\mathcal{M}$ for the same multiphase solid, where the above condition has been replaced by the relation (14₂) between local and overall inelastic strains.

It is thus apparent that the two expressions are distinct, except perhaps in rigid-plastic solids where $\mathbf{M}_r \rightarrow \mathbf{0}$ and $\mathbf{B}_r \rightarrow \mathbf{I}$. However, they can be simultaneously satisfied if the instantaneous stress concentration factor tensors $\mathcal{B}_r$ is evaluated in the manner described in §4. In contrast, some simple algebra shows that if the $\mathcal{B}_r$ are found from (59), then (61) and (62) provide entirely different estimates of the instantaneous inelastic compliance. Therefore, one must conclude that regardless of their apparent popularity in the micromechanics literature, the self-consistent, Mori–Tanaka or other procedures based on (59) and (61), or their analogues, are not admissible in inelastic analysis of heterogeneous media.

That is not to say that the methods themselves are without merit. For example, they may serve in the evaluation of the transformation concentration factor tensors $\mathbf{D}_{sr}$ and $\mathbf{F}_{sr}$ that enter the governing equations (24) and (25), and thus be of use in the analysis outlined in §§4–6. Such evaluation was described in §6 of (I), with the result (I 58) and (I 59) which is reproduced here for completeness:

$$\begin{aligned} \mathbf{D}_{sr} &= [\delta_{sr} \mathbf{I} - c_r (\mathbf{L}^* + \mathbf{L}_s)^{-1} (\mathbf{L}^* + \mathbf{L})] (\mathbf{L}^* + \mathbf{L}_r)^{-1} \mathbf{L}_r \\ &= (\mathbf{I} - \mathbf{A}_s) (\mathbf{L}_s - \mathbf{L})^{-1} (\delta_{sr} \mathbf{I} - c_r \mathbf{A}_r^T) \mathbf{L}_r, \end{aligned} \quad (63)$$

$$\begin{aligned} \mathbf{F}_{sr} &= [\delta_{sr} \mathbf{I} - c_r (\mathbf{M}^* + \mathbf{M}_s)^{-1} (\mathbf{M}^* + \mathbf{M})] (\mathbf{M}^* + \mathbf{M}_r)^{-1} \mathbf{M}_r \\ &= (\mathbf{I} - \mathbf{B}_s) (\mathbf{M}_s - \mathbf{M})^{-1} (\delta_{sr} \mathbf{I} - c_r \mathbf{B}_r^T) \mathbf{M}_r, \end{aligned} \quad (64)$$

where δ_{sr} is the Kronecker symbol, but no summation is indicated by repeated subscripts.

These expressions satisfy the conditions (I 47) to (I 50) which guarantee that the solutions of inelastic problems found from (24) or (25) agree with both (14) and (15) or (16). In the particular case of an elastic–plastic composite, such result may be written as (33₂) or (58₂), and the instantaneous stress concentration factors $\mathcal{B}_r$ found

in this particular way then satisfy both (61) and (62). Note that in contrast to (59), the evaluation of (63) or (64) involves only the constraint tensors and/or the mechanical concentration factor tensors of the elastic composite.

To illustrate these aspects of the problem more completely, we focus on a two-phase elastic-plastic system, $r = \alpha, \beta$, where both phases may experience inelastic deformation. In this case, there are two unknown tensors $\mathcal{B}_r$ which must satisfy the additional relation $c_\alpha \mathcal{B}_\alpha + c_\beta \mathcal{B}_\beta = \mathbf{I}$, so that only one of the unknowns needs to be found. One possible way is to solve (61) and (62) for a single $\mathcal{B}_r$, as

$$\mathcal{B}_\alpha = (1/c_\alpha) [\mathbf{M}_\alpha - \mathbf{M}_\beta - (\mathbf{B}_\alpha^T - \mathbf{I})(\mathcal{M}_\alpha - \mathbf{M}_\alpha) + (\mathbf{B}_\beta^T - \mathbf{I})(\mathcal{M}_\beta - \mathbf{M}_\beta)]^{-1} \times [\mathbf{M} - \mathbf{M}_\beta + (\mathbf{B}_\beta^T - \mathbf{I})(\mathcal{M}_\beta - \mathbf{M}_\beta)], \quad (65)$$

which may be contrasted with (59₂).

To prove that this result is identical to that derived by the analysis of §4, one may use the notation $\mathcal{M}_r^p = (\mathcal{M}_r - \mathbf{M}_r)$ from (30), and then compare (65) with the analogous expression (38₁). First, it is useful to recall that for $r = \alpha, \beta$, the $\mathbf{F}_{r\alpha}$, $\mathbf{F}_{r\beta}$ tensors in (38₁) appear in (39), and that their substitution into (38₁) provides

$$\mathcal{B}_\alpha = [\mathbf{I} - (c_\beta/c_\alpha)(\mathbf{I} - \mathbf{B}_\beta)(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\alpha^p + (c_\alpha/c_\beta)(\mathbf{I} - \mathbf{B}_\alpha)(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\beta^p]^{-1} \times [\mathbf{B}_\alpha + (1/c_\beta)(\mathbf{I} - \mathbf{B}_\alpha)(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\beta^p]. \quad (66)$$

Then, one can proceed as indicated in the Appendix to show that (65) and (66) are indeed identical and satisfy the identity relation preceding (65).

8. Closure

Equations similar to (14) have been known for many years (Levin 1967; Rice 1970; Hill 1971). However, except for the recent papers by Hill (1984, 1985), they have not been widely appreciated in inelastic analysis of heterogeneous solids. One possible reason was the difficulty inherent in evaluation of the residual local fields, which has now been resolved by Dvorak (1990) for two-phase media, and by Dvorak & Benveniste (1992) for multiphase media. This provides a more consistent basis for the theory, but it also modifies certain aspects of the approach that follows (59) and (62). In particular, only elastic solutions of inclusion problems are used in the evaluation of local strains, hence the pronounced directional weakness in constraint of an already yielded aggregate (Hill 1965) is no longer reflected through the solution of an inclusion problem in an elastic homogeneous medium with the instantaneous overall stiffness $\mathcal{L}$, which leads to (59). Instead, this feature is accounted for by solving the inclusion problem in an elastic medium $\mathbf{L}$, which contains the eigenstrain equal to the total overall plastic strain, cf. §6 in (I).

Of course, the absence of inelastic inclusion problems in the analysis provides for a much simpler implementation of specific inelastic constitutive relations into the governing equations (34) or (25); this is illustrated for example by the closed-form expressions (37) to (40) for the elastic plastic solids. That may be particularly advantageous in unit-cell models, which have to use specific inelastic finite element routines. It is now sufficient to use only elastic finite element solutions to find the constant transformation factor tensors in (24) and (25), and then solve these equations directly with any chosen constitutive relations. We will show elsewhere that more efficient procedures can result, especially with coarsely subdivided unit cells that are used in analysis and design of composite structures.

It should be noted in passing that the present results may resolve some questions that are discussed from time to time in the technical literature. One pertains to the possibility of separation of the mechanical and thermal load effects in inelastic multiphase media. The results of §§4–6 make it clear that under combined thermomechanical loading of the aggregate, the temperature change may contribute to the current magnitude of $\sigma_r^{\text{re}}(\mathbf{x})$ and $\epsilon_r^{\text{in}}(\mathbf{x})$, where the relative thermal and mechanical contributions to these total local fields depend on the inelastic response of the phases, the microstructural geometry, and on the combined thermomechanical loading path leading to the current overall magnitudes of σ or ϵ , and θ . Therefore, the thermal contributions to the total inelastic fields may not be separated *a posteriori*, and it may not be possible to define the total thermal strain or relaxation stress, either in (1), or in (5). Only if such definitions can be made for the respective increments, e.g. in the elastic–plastic systems in §4, the total thermal strain may be separated in integration along the path. However, even in the elastic–plastic systems one cannot confirm an inelastic analog of the Levin formula (13₁), although it is well known that the thermal response of certain two-phase elastic–plastic composites can be simulated by mechanical loading along a modified path (Dvorak 1986, 1992).

Another observation can be useful in formulation of so-called inverse problems, which attempt to determine the local inelastic strains from observed changes in surface displacements, or in total overall strains. Note that equations (12) to (14) suggest that any number of local inelastic fields may exist in a heterogeneous medium such that the overall inelastic strain or relaxation stress both vanish. Of course, the existence of such fields may bring into question the uniqueness of solutions of the inverse problems.

This work was supported by the Office of Naval Research and by the ONR/DARPA HiTASC program at Rensselaer. Dr Yapa Rajapaske and Dr Steve Fishman served as program monitors.

Appendix

Here we prove that the instantaneous stress concentration factors in (38₁) or (66), and in (65) both satisfy the relation $c_\alpha \mathcal{B}_\alpha + c_\beta \mathcal{B}_\beta = \mathbf{I}$, and are identical.

Rewrite the $\mathcal{B}_\alpha$ in (66) as

$$\mathcal{B}_\alpha^{(1)} = [\mathcal{B}_\alpha^{(11)}]^{-1} \mathcal{B}_\alpha^{(12)}, \quad (\text{A } 1)$$

where

$$\mathcal{B}_\alpha^{(11)} = \mathbf{I} - (c_\beta/c_\alpha) (\mathbf{I} - \mathbf{B}_\beta) (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\alpha^{\text{p}} - (c_\alpha/c_\beta) (\mathbf{I} - \mathbf{B}_\alpha) (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\beta^{\text{p}}, \quad (\text{A } 2)$$

$$\mathcal{B}_\alpha^{(12)} = \mathbf{B}_\alpha + (1/c_\beta) (\mathbf{I} - \mathbf{B}_\alpha) (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} \mathcal{M}_\beta^{\text{p}}. \quad (\text{A } 3)$$

Similarly, write the $\mathcal{B}_\alpha$ in (65) as

$$\mathcal{B}_\alpha^{(2)} = [\mathcal{B}_\alpha^{(21)}]^{-1} \mathcal{B}_\alpha^{(22)}, \quad (\text{A } 4)$$

where

$$\mathcal{B}_\alpha^{(21)} = \mathbf{M}_\alpha - \mathbf{M}_\beta - (\mathbf{B}_\alpha^{\text{T}} - \mathbf{I}) \mathcal{M}_\alpha^{\text{p}} + (\mathbf{B}_\beta^{\text{T}} - \mathbf{I}) \mathcal{M}_\beta^{\text{p}}, \quad (\text{A } 5)$$

$$\mathcal{B}_\alpha^{(22)} = (1/c_\alpha) [\mathbf{M} - \mathbf{M}_\beta + (\mathbf{B}_\beta^{\text{T}} - \mathbf{I}) \mathcal{M}_\beta^{\text{p}}]. \quad (\text{A } 6)$$

The phase subscripts α, β can be exchanged to obtain

$$\mathcal{B}_\beta^{(1)} = [\mathcal{B}_\beta^{(11)}]^{-1} \mathcal{B}_\beta^{(12)}, \quad \mathcal{B}_\beta^{(2)} = [\mathcal{B}_\beta^{(21)}]^{-1} \mathcal{B}_\beta^{(22)}, \quad (\text{A } 7)$$

where

$$\mathcal{B}_\beta^{(11)} = \mathcal{B}_\alpha^{(11)}, \quad \mathcal{B}_\beta^{(21)} = -\mathcal{B}_\alpha^{(21)} \quad (\text{A } 8)$$

and

$$\mathcal{B}_\beta^{(12)} = \mathbf{B}_\beta - (1/c_\alpha)(\mathbf{I} - \mathbf{B}_\beta)(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}\mathcal{M}_\alpha^{\text{p}}, \quad (\text{A } 9)$$

$$\mathcal{B}_\beta^{(22)} = (1/c_\beta)[\mathbf{M} - \mathbf{M}_\alpha + (\mathbf{B}_\alpha^{\text{T}} - \mathbf{I})\mathcal{M}_\alpha^{\text{p}}]. \quad (\text{A } 10)$$

It is now easy to show that

$$c_\alpha \mathcal{B}_\alpha^{(1)} + c_\beta \mathcal{B}_\beta^{(1)} = [\mathcal{B}_\alpha^{(11)}]^{-1}(c_\alpha \mathcal{B}_\alpha^{(12)} + c_\beta \mathcal{B}_\beta^{(12)}) = \mathbf{I}, \quad (\text{A } 11)$$

$$c_\alpha \mathcal{B}_\alpha^{(2)} + c_\beta \mathcal{B}_\beta^{(2)} = [\mathcal{B}_\alpha^{(21)}]^{-1}(c_\alpha \mathcal{B}_\alpha^{(22)} - c_\beta \mathcal{B}_\beta^{(22)}) = \mathbf{I}. \quad (\text{A } 12)$$

To prove that

$$\mathcal{B}_\alpha^{(1)} = \mathcal{B}_\alpha^{(2)}, \quad \mathcal{B}_\beta^{(1)} = \mathcal{B}_\beta^{(2)}, \quad (\text{A } 13)$$

we focus on the first equality, and write it using (A 1) and (A 4) as

$$[\mathcal{B}_\alpha^{(11)}]^{-1}\mathcal{B}_\alpha^{(12)} = [\mathcal{B}_\alpha^{(21)}]^{-1}\mathcal{B}_\alpha^{(22)}, \quad (\text{A } 14)$$

or as

$$\mathcal{B}_\alpha^{(12)}[\mathcal{B}_\alpha^{(22)}]^{-1} = \mathcal{B}_\alpha^{(11)}[\mathcal{B}_\alpha^{(21)}]^{-1}, \quad (\text{A } 15)$$

under the assumption that the inverses exist.

The relation (A 15) can be proved by verifying that

$$\mathcal{B}_\alpha^{(12)} = (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}\mathcal{B}_\alpha^{(22)}, \quad \mathcal{B}_\alpha^{(11)} = (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}\mathcal{B}_\alpha^{(21)}. \quad (\text{A } 16)$$

This can be easily shown to be true for any finite $\mathcal{M}_\alpha^{\text{p}}, \mathcal{M}_\beta^{\text{p}}$, if one recalls the identities that holds for two-phase materials,

$$c_\alpha(\mathbf{I} - \mathbf{B}_\alpha) = -c_\beta(\mathbf{I} - \mathbf{B}_\beta), \quad c_\alpha \mathbf{B}_\alpha = (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}(\mathbf{M} - \mathbf{M}_\beta), \\ c_\alpha \mathbf{B}_\beta = -(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}(\mathbf{M} - \mathbf{M}_\alpha), \quad (\text{A } 17)$$

and appeals to the symmetry relations

$$\left. \begin{aligned} \mathbf{M} &= \mathbf{M}^{\text{T}}, \quad \mathbf{M}_\alpha = \mathbf{M}_\alpha^{\text{T}}, \quad \mathbf{M}_\beta = \mathbf{M}_\beta^{\text{T}}, \\ \mathbf{B}_\alpha(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} &= (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}\mathbf{B}_\alpha^{\text{T}}, \quad \mathbf{B}_\beta(\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1} = (\mathbf{M}_\alpha - \mathbf{M}_\beta)^{-1}\mathbf{B}_\beta^{\text{T}}. \end{aligned} \right\} \quad (\text{A } 18)$$

Q.E.D.

References

- Benveniste, Y., Dvorak, G. J. & Chen, T. 1991 *J. Mech. Phys. Solids* **39**, 927–946.
- Chaboche, J. L. 1989 *Int. J. Plasticity* **5**, 247–302.
- Chen, T., Dvorak, G. J. & Benveniste, Y. 1992 *J. appl. Mech.* (In the press.)
- Christensen, R. M. 1971 *Theory of viscoelasticity, an introduction*. New York: Academic Press.
- Christensen, R. M. 1979 *Mechanics of composite materials*. New York: John Wiley & Sons.
- Dvorak, G. J. 1986 *J. appl. Mech.* **53**, 737–742.
- Dvorak, G. J. 1990 *Proc. R. Soc. Lond. A* **431**, 89–110.
- Dvorak, G. J. 1991 *Metal matrix composites: mechanisms and properties* (ed. R. K. Everett & R. J. Arsenault), pp. 1–77. Boston: Academic Press.
- Dvorak, G. J. 1992 *J. Thermal Stresses*. (In the press.)
- Dvorak, G. J. & Benveniste, Y. 1992 *Proc. R. Soc. Lond. A* **437**, 291–310. (Preceding paper).
- Gavazzi, A. C. & Lagoudas, D. C. 1990 *Computational Mech.* **7**, 13–19.
- Hill, R. 1965 *J. Mech. Phys. Solids* **13**, 89–101.
- Hill, R. 1966 *J. Mech. Phys. Solids* **14**, 95–102.
- Hill, R. 1967 *J. Mech. Phys. Solids* **15**, 79–95.
- Hill, R. 1971 *Prikl. Mat. Mekh. (PMM)* **35**, 31–39.
- Hill, R. 1984 *Math. Proc. Camb. phil. Soc.* **95**, 481–494.

- Hill, R. 1985 *Math. Proc. Camb. phil. Soc.* **98**, 579–590.
- Hutchinson, J. W. 1970 *Proc. R. Soc. Lond. A* **319**, 247–272.
- Krempel, E. & Lu, H. 1984 *ASME J. Engng Mater. Technol.* **106**, 376–382.
- Levin, V. M. 1967 *Izv. ANSSSR, Mekhanika Tverdogo Tela* **2**, 88–94.
- Mori, T. & Tanaka, K. 1973 *Acta metal.* **21**, 571–574.
- Rice, J. R. 1970 *J. appl. Mech.* **37**, 728.
- Tandon, G. P. & Weng, G. J. 1988 *J. appl. Mech.* **55**, 126–135.

Received 5 August 1991; accepted 5 December 1991